

CLIFFORD HIERARCHY

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1. WHY CLIFFORD HIERARCHY?

The Clifford hierarchy was introduced in [8] by Gottesman and Chuang as a means to achieve universal quantum computation via teleportation. As we may recall from the lectures, the quantum teleportation, as illustrated by Figure 1, enables Alice to teleport a state $|\psi\rangle$ to Bob with the cost of an ancilla Bell state $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. In particular, the correction needed to correct the error in the circuit are classically controlled X and Z gates, which is an element of the Pauli group of two qubits.

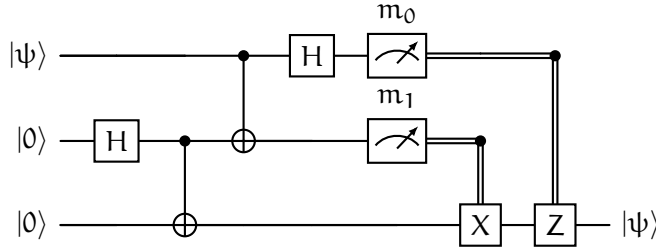


FIGURE 1. Quantum teleportation.

To implement a single-qubit U gate, the circuit proposed in [8], illustrated in Figure 2, prepares the state $(I \otimes U)|\Phi^+\rangle$ and takes a classically controlled gate to correct an error $UPU^\dagger$, where P is a Pauli operator. More precisely, suppose that there is an error P , i.e., the resulting state is $UP|\psi\rangle$ before the error correction, then

$$UP|\psi\rangle = UP(U^\dagger U)|\psi\rangle = (UPU^\dagger)U|\psi\rangle,$$

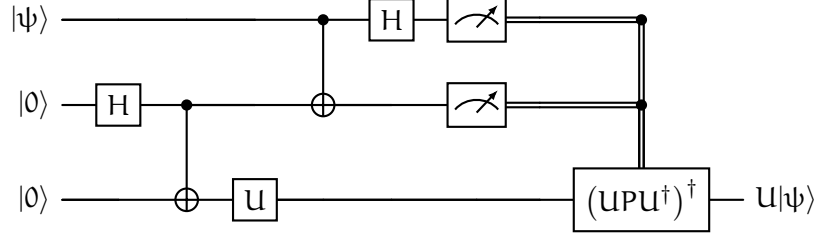
which suggests that we obtain $U|\psi\rangle$ by correcting the error $UPU^\dagger$. Therefore, it is natural to investigate the structure of conjugations of P , which motivates the definition of the Clifford hierarchy.

Definition 1.1. Let $\mathcal{C}_1 = \mathcal{P}_n$ denote the Pauli group on n -qubits. The *Clifford hierarchy* is the sequence of sets recursively defined by

$$\mathcal{C}_{k+1} = \{U \in \mathbb{U}(2^n) : U\mathcal{P}_n U^\dagger \subseteq \mathcal{C}_k\}.$$

Heuristically, an error at the element of $\mathcal{C}_{k+1}$ could be corrected by $\mathcal{C}_k$, and such a process could be done recursively.

For $k = 2$, the set $\mathcal{C}_2$ is the normalizer group of the Pauli group, known as the Clifford group. The difficulty arises when $k \geq 3$, where $\mathcal{C}_k$ is no longer a group. The overarching problem is how to understand these sophisticated sets.

FIGURE 2. Teleportation through the gate U .

2. NOW AND THEN

Due to its extremely complicated structure, there are still a lot to be explored about the Clifford hierarchy. We present below, in chronological order, some important results on this quest. Note that we have chosen to focus on works studying structures of $\mathcal{C}_k$ themselves, and hence do not cover works investigating algebraic structures generated by $\mathcal{C}_k$'s, such as [1].

2.1. semi-Clifford and generalized semi-Clifford gates. The notion of semi-Clifford and generalized semi-Clifford gates are proposed in foundational paper [19] based on previous observations from [20]. The motivation to defining semi-Clifford gates is practical: As illustrated in their analysis, the number of teleportation steps are necessary, on average, to implement a semi-Clifford gate in on the order of 2^n while the corresponding rate is 4^n for other gates.

Definition 2.1. $U \in \mathbb{U}(2^n)$ is **semi-Clifford** if $U = L_1 D L_2$, for some $L_1, L_2 \in \mathcal{C}_2$ and diagonal D . $U \in \mathbb{U}(2^n)$ is **generalized semi-Clifford** if $U = L_1 \Pi D L_2$, for some $L_1, L_2 \in \mathcal{C}_2$, a permutation matrix Π , and diagonal D .

It is well-known that semi-Clifford and generalized semi-Clifford gates admit a more geometric characterization.

Theorem 2.2. U is semi-Clifford iff $U\mathcal{G}U^\dagger \subseteq \mathcal{P}_n$ is a maximal abelian subgroup for a maximal abelian subgroup $\mathcal{G} \subseteq \mathcal{P}_n$. U is generalized semi-Clifford iff $U \text{span}(\mathcal{G})U^\dagger = \text{span}(\mathcal{G}')$ for maximal abelian subgroups $\mathcal{G}, \mathcal{G}' \subseteq \mathcal{P}_n$.

The relationship between the Clifford hierarchy and these gates are

Theorem 2.3. The follow results hold for Clifford hierarchy

- All gates in $\mathcal{C}_k$ are semi-clifford operations for $(n = 1, 2, \forall k)$ [19].
- All gates in $\mathcal{C}_k$ are semi-Clifford operations for $\{n = 3, k = 3\}$ [19].
- Not all gates in $\mathcal{C}_k$ are semi-Clifford operations [19].
- For $n = 3$, all gate in $\mathcal{C}_3$ are generalized semi-Clifford [4].
- For $n = 4$, all gate in $\mathcal{C}_3$ are generalized semi-Clifford [2].

As one can see, understanding $\mathcal{C}_3$ is already quite laborious. In fact, although it has been known since 2010 that all gate in $\mathcal{C}_3$ are generalized semi-Clifford, it was not until 2025 that it was proved that there exists non-semi-Clifford gates in $\mathcal{C}_3$ in [9, Lemma 2.6].

Throughout this continuous effort, tools from different areas of mathematics have been applied including group theory, symplectic geometry, finite fields, coding theory, and Fourier analysis.

2.2. Diagonal gates in the Clifford hierarchy. In [19], the authors also proposed to study the diagonal gates in the Clifford hierarchy, which retains a group structure.

Theorem 2.4 ([19]). For all k , the set of n -qubit diagonal gates in $\mathcal{C}_k$ forms a group.

As it turns out, these groups not only have nice structure, but can also be explicitly computed. The following result is obtain in [6], where their results are generalized to p -qudits, meaning that each qudit has states $|0\rangle, |1\rangle, \dots, |p-1\rangle$. A qubit is hence the case $p = 2$.

Theorem 2.5 ([6]). Let p be prime. For $\mathbf{j} \in \mathbb{Z}_p^n$ and $\mathbf{a} \in \{0, \dots, p-1\}^n$, let $\mathbf{j}^{\mathbf{a}} = j_1^{a_1} j_2^{a_2} \dots j_n^{a_n}$. Then U is a diagonal gate in $\mathcal{C}_k$ if and only if it can be written as

$$U = e^{i\phi} \sum_{\mathbf{j} \in \mathbb{Z}_p^n} \exp \left(2\pi i \sum_{\substack{m \geq 1, \mathbf{a} \in \{0, \dots, p-1\}^n \setminus \{0\} \\ (p-1)(m-1) + |\mathbf{a}| \leq k}} \frac{c_{m,\mathbf{a}}}{p^m} \mathbf{j}^{\mathbf{a}} \right) |\mathbf{j}\rangle\langle\mathbf{j}|, \quad \phi \in \mathbb{R}, c_{m,\mathbf{a}} \in \mathbb{Z}_{p^m}.$$

2.3. Climbing the hierarchy. For diagonal gates, we can climb the hierarchy by taking square roots, which is always possible for diagonal gates. A simple example is that for $Z \in \mathcal{C}_1$, $\sqrt{Z} = S \in \mathcal{C}_2$ and $\sqrt{S} = T \in \mathcal{C}_3$. The practical implementation of these diagonal climbings can be realized in a CSS code, via a constructive algorithm.

Definition 2.6. A CSS code is a quantum error correcting code that uses entanglement over physical qubits to create a "logical" error corrected qubit. These codes can be described by $[[n, k, d]]$ where n is the number of physical qubits, k is the number of logical qubits, and d is the code distance. The distance of a code says that $d - 1$ errors can be corrected and if $d = 2t - 1$ then t errors can be corrected [15].

The following algorithm allows a CSS code implementing a gate to increase that gates logical level [11]. Starting from a CSS code realizing a logical gate in $\mathcal{C}_\ell$, one first concatenates

$$[[n, k, d]] \longrightarrow [[2n, k, d' \geq d]],$$

which permits a physical gate one level higher while preserving the same logical gate. Removing suitable Z -stabilizers then splits the generator coefficients and can raise the logical gate from $\mathcal{C}_\ell$ to $\mathcal{C}_{\ell+1}$, while adding X -stabilizers can recover lost code distance. Thus,

$$\boxed{\text{concatenate} \rightarrow \text{remove } Z\text{-stabilizers} \rightarrow \text{add } X\text{-stabilizers}}$$

provides a systematic way of constructing CSS codes for higher-level diagonal logical gates.

For a Hermitian unitary U , define

$$\hat{U} = e^{i\pi U/4} = \frac{I + iU}{\sqrt{2}}, \quad \hat{U}^2 = iU,$$

so that $\hat{U}$ is a square root of U up to global phase. Unlike the diagonal case, a square root does *not* always climb the hierarchy (for example, the Hadamard gate does not). For a Hermitian Clifford $C \in \mathcal{C}_2$, [3] characterizes when $\hat{C} \in \mathcal{C}_3$ through the binary symplectic matrix F_C [7] describing its action on Paulis:

$$CE(v)C^\dagger = \pm E(vF_C).$$

The relevant climbing condition is that F_C be hyperbolic and that its residue space

$$\text{Res}(F_C) = \text{rowspan}(I + F_C)$$

have dimension 2. Thus the problem of deciding whether a Clifford square root climbs to level three reduces to a finite-field linear-algebra test.

2.4. Permutation gates in the third level of the Clifford hierarchy. Adding non-Clifford gate to the Clifford group forms a universal gate set. This paper [9] studies gates in the third level of the Clifford hierarchy $\mathcal{C}_3$ and the permutation gates $\mathcal{C}_3^{\text{sym}}$ (that permutes the 2^n computational basis states). Gates in $\mathcal{C}_3$ can be implemented fault-tolerantly and gates in $\mathcal{C}_3^{\text{sym}}$ can be implemented with low-cost for FTQC.

Definition 2.7 (Staircase form Toffoli circuits). A product of pairwise distinct Toffoli gates is said to be in *staircase form* if each gate $\text{TOF}_{i,j,k}$ in the product has $i < j < k$ and the target qubits are in nondecreasing order in the order that the gates are applied.

The authors first characterize all permutations gates in $\mathcal{C}_3$ could be factored into Clifford operations and a staircase form permutation.

Theorem 2.8. Suppose $\pi \in \mathcal{C}_3$ is a permutation gate. Then there exist Clifford permutations ϕ_1, ϕ_2 and a staircase form permutation $\mu \in \mathcal{C}_3$ such that $\pi = \phi_1 \mu \phi_2$.

More amusingly, the seemingly artificial definition of the staircase form is connected to the algebraic structure of *descending multiplications*: A map $\mathbb{F}_2^n \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is called a descending multiplication if:

- it is linear in each coordinate (distributive), associative, and commutative;
- $e_i e_i = e_i^2 = 0$ for all $i \in [n]$;
- $e_i e_j \in \text{span}\{e_k : k > j\}$ for all $i < j$ in $[n]$.

Theorem 2.9. Every staircase form permutation gate in $\mathcal{C}_3$ induces a descending multiplication over $\mathbb{F}_2^n$. In correspondence, every descending multiplication over $\mathbb{F}_2^n$ induces a staircase form permutation gate in $\mathcal{C}_3$.

Lastly, via the machinery developed along the way to obtain the above results, the authors construct a novel family of permutations in the Clifford hierarchy which disproves two conjectures in [1].

Theorem 2.10. For all $k \geq 3$, we may construct $U_k \in \mathcal{C}_3$ such that $U_k^\dagger \notin \mathcal{C}_k$.

2.5. Logical Clifford synthesis. On the implementation side, the question is: how do we implement logical Clifford (or higher) operators on stabilizer codes by synthesizing physical circuits?

(1). On $\mathcal{C}_2$ there's a clean answer by [16] using the identification

$$\phi : \mathcal{C}_2 / \langle U(1), \text{HW}_N \rangle \xrightarrow{\sim} \text{Sp}_{2n}(\mathbb{F}_2).$$

Writing $E(v) = X^a Z^b$ for $v = (a, b) \in \mathbb{F}_2^{2n}$, every Clifford satisfies $CE(v)C^\dagger = \pm E(vF_C)$, and F_C is symplectic precisely because conjugation preserves commutation. The non-commutativity of HW_N is thus a symplectic form, and $\mathcal{C}_2$ becomes linear algebra over $\mathbb{F}_2$.

The point of [16] is that a logical Clifford does *not* determine an element of $\text{Sp}_{2n}(\mathbb{F}_2)$. Fix an $[[n, k]]$ code with stabilizer S , set $r = n - k$, and choose a symplectic basis of $\mathbb{F}_2^{2n}$ adapted to the code: stabilizers $\hat{S}_1, \dots, \hat{S}_r$, destabilizers $\hat{T}_1, \dots, \hat{T}_r$ with $\langle \hat{T}_i, \hat{S}_j \rangle_s = \delta_{ij}$ and $\langle \hat{T}_i, \hat{T}_j \rangle_s = 0$, and k conjugate logical pairs. A target $\bar{F} \in \text{Sp}_{2k}(\mathbb{F}_2)$ pins the $2k$ logical images and the requirement $\text{CSC}^\dagger = S$ pins the r stabilizer images, leaving the destabilizer images free. What must be completed is a *partial* symplectic matrix.

Theorem 2.11 ([16]). The solutions are enumerated efficiently by symplectic transvections, and there are exactly $2^{r(r+1)/2}$ of them, where $r = n - k$.

This enumeration can actually be completed further. More precisely, the residual freedom is $\hat{T}_i \mapsto \hat{T}_i + \sum_j \Lambda_{ij} \hat{S}_j$, which preserves all pairings except $\langle \hat{T}_i, \hat{T}_j \rangle_s = \Lambda_{ij} + \Lambda_{ji}$, so isotropy forces $\Lambda = \Lambda^T$, and symmetric $r \times r$ matrices over $\mathbb{F}_2$ number $2^{r(r+1)/2}$. All solutions implement the same

logical operator and differ only outside the code space, so one decomposes each into elementary circuits and optimizes the family for depth or two-qubit gate count, correcting signs at the end by a Pauli. This freedom is not the whole gauge group: in [12] the group of physical Cliffords acting trivially on the code space is parameterized rather than enumerated, picking up further factors $|\text{GL}_r(\mathbb{F}_2)|$ from relabelling stabilizer generators and 2^{2kr} from shifting logical representatives by stabilizers, which allows synthesis to be posed as an optimization with device connectivity imposed as a constraint.

(2). What about the fault-tolerant aspect? Neither result involves the distance d since both are statements about S as a subgroup, hence about n and k alone. A synthesized circuit is therefore not fault-tolerant by construction, since an intermediate two-qubit gate may spread a single fault beyond the correction capability; a complementary line of work restricts to gates induced by code automorphisms [18], gaining fault tolerance and losing completeness.

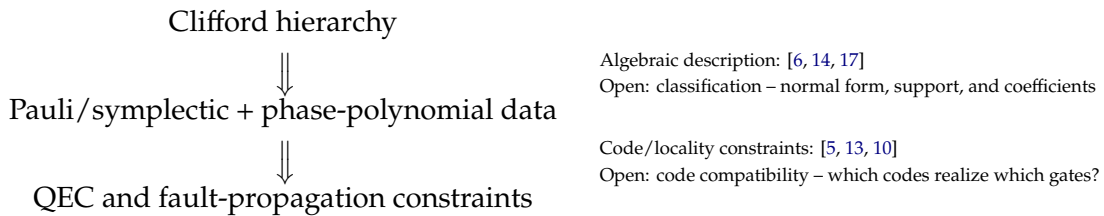
(3). The obstruction above the second level is structural rather than technical. $\mathcal{C}_j$ is not a group for $j \geq 3$, so there is no analogue of ϕ and no linear system to solve. Two substitutes survive. The diagonal elements of $\mathcal{C}_k$ do form a group [19] with the explicit normal form of [6], so diagonal logicals remain tractable, and [11] works in exactly this regime. More generally, if $U = L_1 D L_2$ with $L_1, L_2 \in \mathcal{C}_2$ and D diagonal, synthesizing a logical U reduces to two instances of logical Clifford synthesis together with a diagonal gate; the generalized semi-Clifford conjecture of [19], established at the third level in [4], is therefore not merely structural, since it would make Theorem 2.11 a subroutine for the entire hierarchy up to the permutation factor.

3. OPEN PROBLEMS

As illustrated in our previous discussions, there are still a lot to be understood about characterizing the Clifford hierarchy.

- (1) Are all gate in $\mathcal{C}_k$ generalized semi-Clifford for $k \geq 4$?
- (2) For $n = 5, 6$, is there a $\mathcal{C}_3$ operator that is non-semi-Clifford?
- (3) Is there a more general algebraic characterization of $\mathcal{C}_3$?
- (4) Besides semi-Clifford, generalized semi-Clifford, permutation, and diagonal, what are some other structured operators that can help us study the Clifford hierarchy?

A promising goal is to connect three descriptions in one framework:



Such a bridge would turn structural results about $\mathcal{C}_3$ and higher levels into constructive recipes for protected logical gates.

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